

1. Use Case: Place Order

Goal: Allow customers to purchase products in their cart after authentication.

- **Scenario: Place order successfully**
- **Given** the customer is logged in
- **And** the cart has at least one product
- **When** the customer confirms checkout
- **Then** the system creates a new order in the database
- **And** the system clears all items in the customer's cart (BR-06)
- **And** the system displays the "Order Confirmation" message with Order ID
- **Scenario: Checkout with empty cart**
- **Given** the customer is logged in
- **And** the cart is empty
- **When** the customer clicks checkout
- **Then** the system displays an "Empty cart" message (BR-04)
- **And** the system prevents the transition to the Shipping Information page

2. Use Case: Authentication (Login)

Goal: Verify customer credentials before allowing sensitive actions.

- **Scenario: Login successfully**
- **Given** the customer has a valid account in the system
- **When** the customer enters the correct username and password
- **Then** the system authenticates the user and redirects to the previous page or homepage
- **Scenario: Login failed**
- **Given** the customer enters an incorrect username or password
- **When** the customer submits the login form
- **Then** the system displays the error message: "Invalid username or password"
- **And** the customer remains on the Login page

3. Use Case: Admin – Manage Order

Goal: Allow administrators to control the order fulfillment lifecycle.

- **Scenario: Update order status**
- **Given** the admin is logged in with administrative privileges
- **When** the admin updates the status of an order to "Shipped"
- **Then** the system updates the order status successfully in the database
- **And** the system logs the timestamp of the status change
- **Scenario: Prevent cancellation of Shipped orders**

- **Given** an order has the status "Shipped"
- **When** any user (Admin/Customer) attempts to click "Cancel Order" (BR-08)
- **Then** the system disables the Cancel button or displays a message: "Shipped orders cannot be cancelled"

4. Business Validation Rules (Summary for Testers)

- **Inventory Check:** System must validate stock availability before finalizing the order.
- **Price Integrity:** The unit price in OrderItems must be frozen at the time of confirmation, regardless of future product price changes (BR-09).
- **Data Security:** Admin cannot view customer passwords in plain text (Security Best Practice).